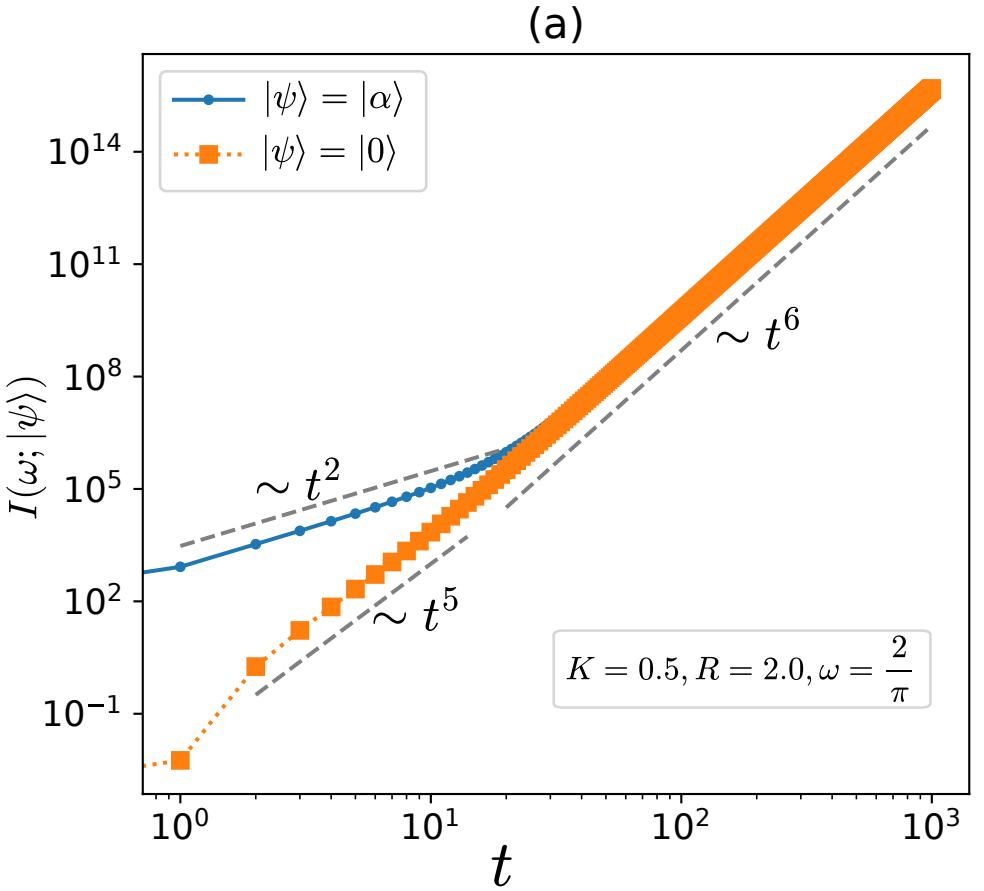

(a)
$|\psi\rangle = |\alpha\rangle$
$|\psi\rangle = |0\rangle$
$\sim t^2$
$\sim t^5$
$\sim t^6$
$K = 0.5, R = 2.0, \omega = \frac{2}{\pi}$
$I(\omega; |\psi\rangle)$
t
10^{14}
10^{11}
10^{8}
10^{5}
10^{2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}